

Week 1

Date: Dec. 15, 2024

Task Description:

Analysis of current dataset recommender systems and identification of loopholes.

Study of current multilingual embedding models: Sentence-BERT, Hugging Face Transformers.

Training open multilingual evaluation and testing datasets.

Time Spent: 6 h

Special Problems Faced or Difficulties Encountered:

Hard to find multicultural data entries that have compatible metadata.

Grasping the idea of embedding alignment in different languages.

Action: Solution or Action Taken:

Kaggle and Hugging Face Datasets as a platform to search and collect required datasets of all languages.

Reading through research papers that discuss cross-lingual embedding approaches to context alignment.

Learning Outcomes:

Better grasp of embedding-based models and finishes learning strategies for pre-processing.

Week 2

Date: 22/12/2024

Task Description:

Engineered the architecture of a multilingual recommendation engine.

Developed a pre-processing pipeline for user queries in multiple languages.

Experimented with sentence embeddings for similarity scoring.

Total number of hours spent: 8 h

Challenges and Difficulties Encountered:

Differing nuances cause language-specific interpretations of queries.

Inconsistent performance across languages.

Solution or Action Taken:

Normalization of queries that are language-specific.

Fine-tuning of pre-trained multilingual models of embeddings.

Learning Outcomes:

Better handling preprocessing and normalization of queries.

Experience with fine-tuning an embedding model in the real world for practice by doing so.

Week 3

Date: Dec. 29, 2024

Task Description:

Integration of the recommendation engine in vector DB RAG operations.
Evaluation of system performance with test queries in Hindi, English, Malayalam, and Kannada.

Metrics like precision and recall visualize recommendatory accuracy.

Time Spent: 9 hours.

Challenges of Difficulties Encountered:

Speed along with accuracy at real-time query processing.

Data-deficient resource languages like Malayalam yield an insufficient amount of data for the databases.

Actions Taken or Solutions to the Problems:

Optimized similarity computation of embeddings for an efficient process.

Created more datasets and synthetic data to add robustness to the system.

Learning Outcomes:

Understanding how to optimize recommendation systems for real-time use.

Ways of handling problems with low-resourced languages learned.

Weekly Summary:

Total time spent: 23 hours.

Key achievements:

Designed and implemented the initial prototype of the dataset recommendation engine.

Slightly productive in dealing with challenges in multilingual embedding alignment and query interpretation.

A pipeline set up to continuously evaluate and improve recommendation accuracy.